

B.Tech. Degree I Semester Examination November 2019

CS/EC/IT 19-200-0103 B ENGINEERING MECHANICS
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

PART A (Answer ALL questions)

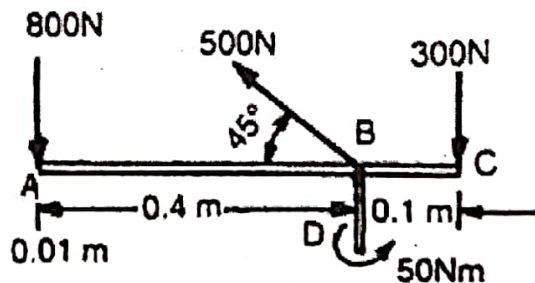
(8 × 3 = 24)

- I. (a) State and prove Varignon's Principle.
- (b) Explain about equivalent force system.
- (c) State and prove Pappus Theorems.
- (d) Derive the equation for product of inertia of a quarter of a circle.
- (e) Derive three equations of motion.
- (f) Explain stable and unstable equilibrium with the help of examples.
- (g) Explain and derive the principle of angular momentum.
- (h) Describe about compound pendulum.

PART B

(4 × 12 = 48)

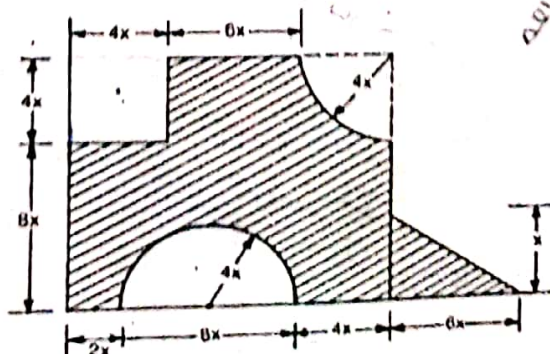
- II. (a) A bracket is subjected to a coplanar force system as shown in figure. (6)
Determine the magnitude and line of action of the single resultant force of the system at point A and find the distance of its line of action from the point B.



- (b) A 7m long ladder rests against a wall with which it makes an angle of 45° and on a floor. If a man whose weight is one half of the ladder, climbs it at what distance along the ladder will he be when the ladder is about to slip? Coefficient of friction between the ladder and the wall is $\frac{1}{3}$ and that between the ladder and floor is $\frac{1}{2}$. (6)

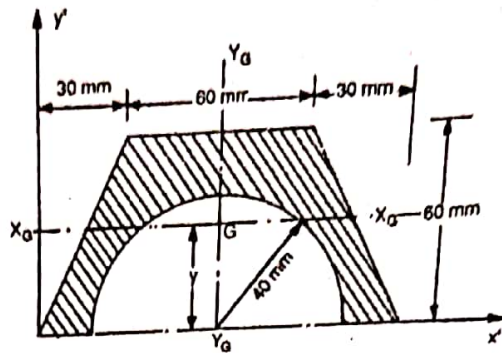
OR

Determine the centroid of the complex area as shown in figure. (12)



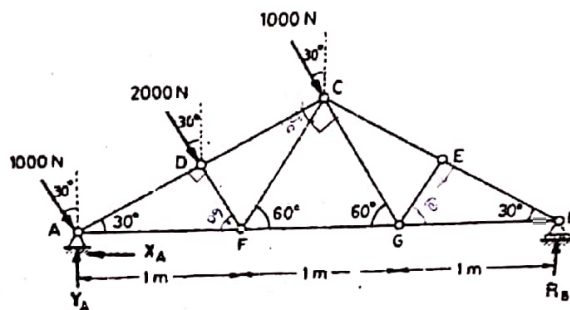
(P.T.O.)

- IV. Compute the moment of inertia of composite area shown in figure about centroidal axes. Also determine the radius of gyration of the area about X axis through the centroid. (12)

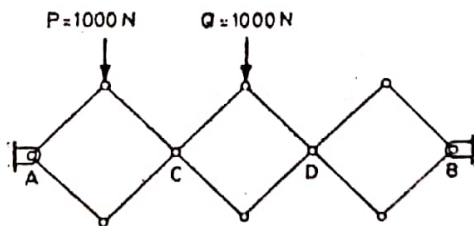


OR

Analyse the truss shown in figure. (12)

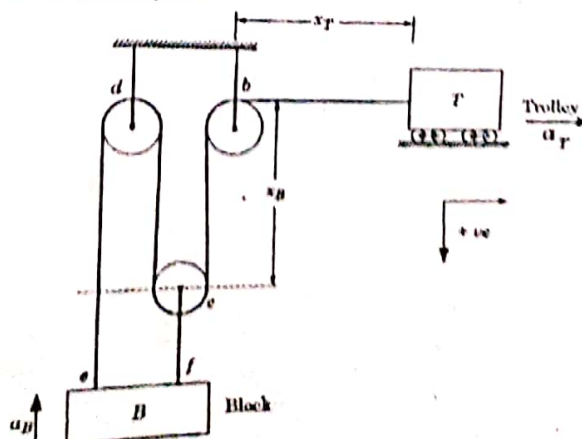


Determine the horizontal and vertical components of the reactions at A and B of the frame which is made up of three squares with hinged joints as shown in figure, by applying the principle of virtual work concept. (12)



OR

- VII. (a) A trolley resting on a horizontal plane starts from rest and is moved to the right with constant acceleration of 0.18 m/s^2 . Determine the (i) acceleration of block B (ii) velocities of the trolley and the block after a time 4 seconds and the distance moved by them. (6)

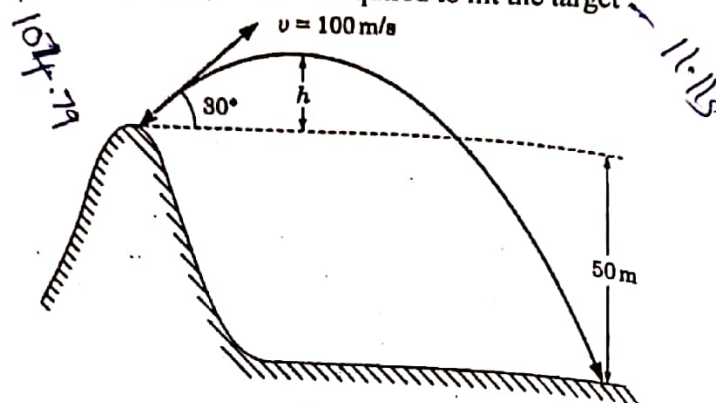


(Contd...3)

- (b) A particle is performing SHM. When it is at a distance of 10cm and 20cm from mean position, its velocities are 1.2 m/s and 0.8 m/s respectively. Find (i) the amplitude of oscillation (ii) the time period of oscillations (iii) the maximum velocity (iv) its maximum acceleration. (6)

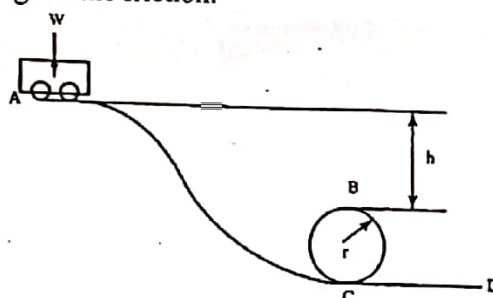
VIII.

- A soldier positioned on a hill fires a bullet at an angle of 30° upwards from the horizontal. The target lies 50m below him and the bullet is fired with a velocity of 100m/s. Determine (i) the maximum height to which the bullet will rise above the position of the soldier, (ii) the velocity with which the bullet will hit the target (iii) the time required to hit the target (12)

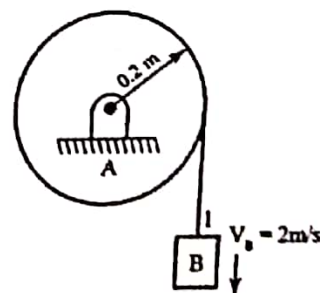


OR

- IX. (a) A small car of weight W starts from rest at A and rolls without friction along the loop ACBD. What is the least height h above the top of the loop at which the car can start without failing off the track at B and for such a starting position what velocity will the car have along the horizontal portion CD of the track. Neglect the friction. (6)



- (b) A block of weight 59N attached to a cord which is wrapped around the periphery of a disc of weight 196N and radius 0.2m which has a moment of inertia 0.4 kgm^2 about the axis of rotation. If the block is initially moving downwards with a speed of 2m/s, determine the speed in 3 seconds, neglecting the mass of the string. (6)



$$2.4 \quad 2 \times 100 \times \frac{1}{2} \\ \frac{1}{2} \times 9.8$$

$$\frac{9.8}{1000}$$